Roddneil B. Gemina

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PROFESSIONAL SUMMARY

Computer Science graduate from CIT-U with hands-on expertise in full-stack web applications, responsive user interfaces, and scalable backend services. Demonstrated ability delivering high-reliability systems, verified through industry internships at Opswerks (SRE) and DXForm (QA), alongside international research recognition. Eager to contribute scalable, test-driven code to Foundry for Good as Software Engineer.

EDUCATION

Bachelor of Science in Computer Science — Cebu Institute of Technology - University (*2022 - 2026* | *Cebu City, Philippines*)

- Presidential OIC & Bass 2 - Koro Teknoy Chorale
- Scopus-Indexed Conference Paper Author (ICETT 2026)
- Philippine Computing Science Congress Presenter (PCSC 2026, Davao)

Secondary Education (STEM Strand) — Philippine Science High School - Central Visayas Campus (*2016 - 2022* | *Argao, Cebu, Philippines*)

- Graduated With Honors
- Dean's Lister Placement
- Red Hot Pawns Chess Club (2 Years)

INDUSTRY CERTIFICATIONS & HONORS

- **PhilNITS FE (Fundamental Engineer)** (Philippine National IT Standards Foundation, April 2025): Comprehensive engineering certification assessing discrete structures, algorithms, databases, networking, and software design.
- **TOPCIT Level 3 Expertise** (Test of Practical Competency in IT (TOPCIT), 2025): Demonstrated advanced proficiency in software development, technical communication, and systems architecture.
- **AWS Academy Graduate - Cloud Architecting & Cloud Foundations** (Amazon Web Services (AWS), 2024 - 2025): Architecting multi-tier cloud solutions, VPC networking, IAM security, and resilient microservices on AWS.
- **CodeChum Certified in Java & C Programming** (CodeChum, 2023 - 2024): Demonstrated mastery in object-oriented programming, data structures, and memory management.

WORK EXPERIENCE

Site Reliability Engineering (SRE) Intern | Opswerks (2026 — Cebu City, Philippines)

- Administered Linux server environments, configuring AWS cloud infrastructure, Docker containers, and Kubernetes pods.
- Built automated deployment pipelines and integrated telemetry and monitoring tools to maintain high application availability.
- Automated repetitive operational workflows using Python and Bash scripting, reducing manual incident troubleshooting time.

Quality Assurance (QA) Analyst Intern | DXForm Technologies (2025 — Cebu City, Philippines)

- Completed 250 hours of On-the-Job Training (OJT), managing test strategy execution across 4 client web and mobile applications.
- Authored structured test plans, executed regression and smoke suites, and logged high-impact defects with reproducible steps in Jira.
- Collaborated closely with software engineering teams to validate bug fixes and enhance cross-browser compatibility standards.

AI & Software Researcher | CIT-U College of Computer Studies (2025 - 2026 — Cebu City, Philippines)

- Primary author of 'Fusion Rush: Enhancing Proficiency in Boolean Logic Algebra through Gamification', accepted at Scopus-indexed ICETT 2026.
- Developed and presented 'Visual AI Logic Proof Parser & Evaluator' at the Philippine Computing Science Congress (PCSC 2026, Davao).
- Engineered rule-based inference engines and computer vision OCR pipelines to evaluate handwritten mathematical logic proofs.

FEATURED PROJECTS & RESEARCH

Fusion Rush: Gamified Boolean Logic & AI Tutoring (2025 - 2026) — *Tech: Python, JavaScript, Gamification Engine, Discrete Math*

- Educational AI tutoring software providing gamified learning on the Rules of Inference for Discrete Math. Tested with CIT-U CS students and accepted at the Scopus-indexed ICETT 2026 Conference.
- Built a modular inference verification engine in Python to validate multi-step logic deductions in real-time.
- Designed interactive user interfaces reducing student error rates on symbolic logic exercises by over 30%.

Visual AI Logic Proof Parser & Evaluator (2026) — *Tech: Python, Visual AI, OCR, Computer Vision, Logic Verification*

- System using Visual AI to scan, parse, and evaluate validity and score handwritten Discrete Math proofs. Presented at PCSC 2026 in Davao.
- Created an end-to-end OCR and structural parsing pipeline converting handwritten proofs into structured ASTs.
- Automated step-by-step mathematical soundness verification, delivering instant pedagogical feedback.

TakeIt — Event Management & Ticketing Platform (2024) — *Tech: React, Node.js, Express, PostgreSQL, REST APIs*

- Scalable event booking platform providing smooth ticketing workflows, seat selection, organizer dashboards, and QR ticket issuance.
- Engineered responsive React frontend integrated with RESTful Node.js backend services and PostgreSQL schemas.

- Implemented real-time ticket inventory locking mechanisms to prevent duplicate bookings under concurrent traffic.

Nexchef — Culinary Platform with Synchronized Timers (2024 - 2025) — *Tech: React, TypeScript, Node.js, Supabase, WebSockets*

- Interactive recipe and culinary web application featuring live multi-stage recipe timing and community recipe sharing.
- Architected Supabase backend integration for real-time user session sync and state persistence.
- Designed clean component hierarchy with optimized state management.

Bomberman BattleRoyale (2023) — *Tech: Java, Multi-threading, TCP/IP Sockets, Swing*

- Multiplayer arcade battle royale game built in Java featuring low-latency socket networking and thread synchronization.
- Implemented multi-threaded socket communication supporting synchronized concurrent player state updates.
- Engineered collision detection, map destruction, and grid mechanics from scratch in Java.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, JavaScript (ES6+), TypeScript, PHP, SQL, HTML5, CSS3, Bash
Frameworks: React, Next.js, Node.js, Express, Django, FastAPI, Android Studio
Cloud Devops: AWS (EC2, S3, VPC, IAM), Kubernetes, Docker, Linux Administration, CI/CD, Supabase, Git
Databases: PostgreSQL, MySQL, SQLite, Supabase
Testing Tools: Quality Assurance, Jira, Automated Testing, Postman, Jest
Core Competencies: Data Structures & Algorithms, Discrete Mathematics, Systems Architecture, Site Reliability Engineering, Visual AI & OCR